JGi JAIN DEEMED-TO-BE UNIVERSITY | FACULTY OF ENGINEERING AND TECHNOLOGY

PROJECT-CENTRIC LEARNING (PCL) – PCL - 41

A CONTEXT-AWARE REAL-TIME ASSISTIVE VISION SYSTEM WITH PRIORITY-BASED AUDIO GUIDANCE FOR THE VISUALLY IMPAIRED : VOCAL EYES

Review Presentation – 3

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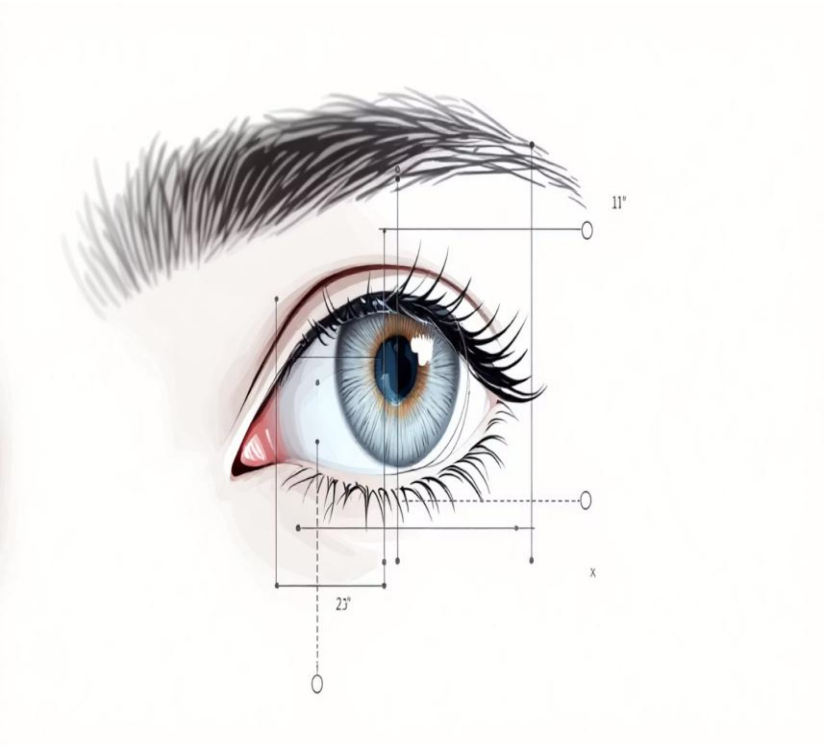
Abstract

- Visual impairment limits safe navigation and environmental awareness, while existing assistive systems rely on continuous narration without safety prioritization.
- VocalEyes is a real-time AI-based assistive vision system that uses YOLOv8 for object detection, spatial awareness, and priority-based alerts. It incorporates intelligent silence to reduce repetitive audio and safety-gated OCR for controlled text reading.
- The system converts live camera input into meaningful audio guidance, supports offline text-to-speech, and includes voice control and an interactive dashboard. It improves safety, reduces cognitive overload, and is suitable for real-world deployment on low-cost devices.

Introduction

- Visually impaired individuals lack real-time perception of surroundings
- Traditional assistive tools provide limited feedback
- Current AI applications narrate continuously without safety awareness
- Absence of spatial understanding increases navigation risk
- Unlike existing systems, VocalEyes integrates decision intelligence, user interaction, and voice-based control into a single assistive framework.



- Our primary motivation is to build a decision-aware assistive system that prioritizes user safety and intelligently manages audio output to reduce unnecessary narration, making navigation more intuitive and less overwhelming.

- To translate theoretical AI concepts into a practical assistive solution that can improve everyday mobility and independence for visually impaired users.

System Requirements

1. **Hardware Requirements:**

- Laptop / Desktop with minimum Intel i5 or equivalent processor
- Minimum 8 GB RAM (16 GB recommended for smoother YOLO inference)
- Integrated or dedicated GPU (optional, CPU execution supported)
- Webcam or external USB camera for live video input
- Speaker or earphones for audio output

2. **Software Requirements**

- Operating System: Windows
- Programming Language: Python 3.10+
- Object Detection Framework: YOLOv8 (Ultralytics)
- Computer Vision Library: OpenCV
- OCR Engine: Tesseract OCR
- Speech Output: Windows Speech API (offline TTS)
- Development Environment: VS Code / PyCharm
- Supporting Libraries: NumPy, PyTorch, Pillow, pytesseract

Literature Survey

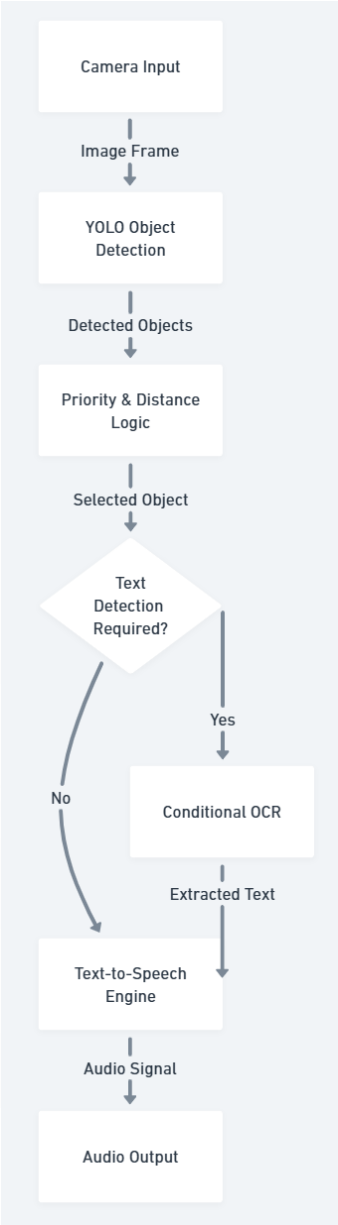
Study	Summary	Critical Evaluation
Seeing AI	Mobile-based assistive application providing object recognition, scene description, and OCR-based text reading using computer vision.	Continuous narration without distance or direction awareness; lacks priority-based safety alerts, leading to cognitive overload.
Envision Glasses	Wearable smart glasses integrating OCR, object recognition, and face identification for hands-free assistance.	Proprietary and high-cost solution; absence of transparent decision logic; no time-based intelligent silence or risk-based alert mechanism.
Be My Eyes	Provides visual assistance through human volunteers or AI-based scene description via smartphones.	Requires internet connectivity; not autonomous; does not support real-time obstacle detection or navigation safety.

OrCam MyEye	Embedded wearable device enabling text reading and object identification.	Closed-source architecture with limited customization; expensive hardware; lacks spatial estimation and contextual prioritization.
YOLO-Based Assistive Prototypes	Utilize deep learning models for real-time object detection with text-to-speech output.	Primarily follow a “detect and speak” approach; no context awareness, distance estimation, or time-based intelligent silence.
OCR-Based Reading Systems	Apply optical character recognition to extract printed text and convert it to speech.	OCR operates continuously without safety context, potentially distracting users during hazardous situations.
Traditional Assistive Tools	Employ canes or ultrasonic sensors to detect nearby obstacles.	Provide only proximity alerts; lack semantic understanding, object classification, and environmental context.

Existing System

- Current assistive vision systems mainly provide object detection and basic OCR-based text reading. However, most operate as simple narrators and lack decision intelligence, safety prioritization, spatial awareness, and intelligent silence.
- **Commercial Devices :** Envision Glasses offer object recognition and OCR but are closed-source, expensive, and lack priority-based alerts or adaptive audio control.
- **Mobile Applications :** Seeing AI provides scene narration and text reading, but speaks continuously without distance awareness or safety classification.
- **Human-Assisted Platforms :** Be My Eyes depends on internet-connected volunteers and does not support autonomous real-time obstacle detection.
- **Wearable OCR Devices :** OrCam MyEye enables text reading but is proprietary, costly, and lacks customizable decision logic.

Block Diagram



Proposed System

- VocalEyes is a context-aware assistive vision system designed to provide intelligent audio guidance for visually impaired users by combining real-time object detection, spatial awareness, and safety-driven decision logic.

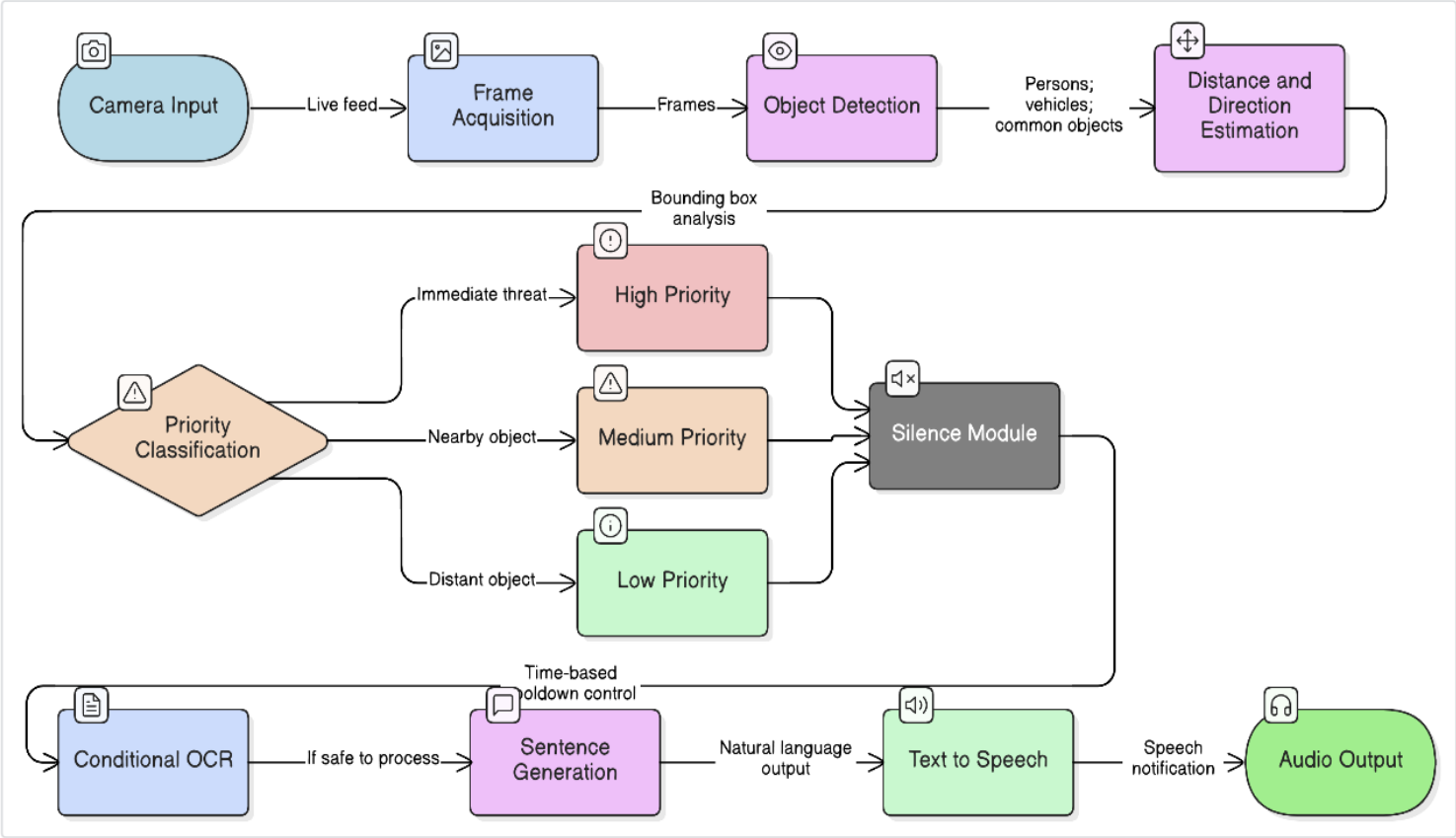
Key Components

- Real-time object detection (YOLO + live camera)
- Distance & direction estimation (bounding boxes)
- Priority-based safety alerts
- Intelligent silence for reduced repetition
- Safety-controlled OCR for text reading
- Offline text-to-speech assistance
- Interactive dashboard (object & danger count)
- Detection history tracking
- Voice control (ON/OFF toggle) and speech commands for hands-free interaction

Existing System Vs Proposed System

Feature	Existing Systems	VocalEyes
Object Detection	Yes	Yes
Spatial Awareness	Limited / None	Distance & Direction Estimation
Safety Prioritization	Not implemented	Priority-Based Alerts
Silence Control	Continuous narration	Time-Based Intelligent Silence
OCR	Always active	Safety-Gated OCR
Decision Intelligence	Minimal	Layered Decision Pipeline
Offline Operation	Often internet dependent	Fully Offline
Customization	Closed / Proprietary	Modular Architecture

Proposed System Architecture Diagram

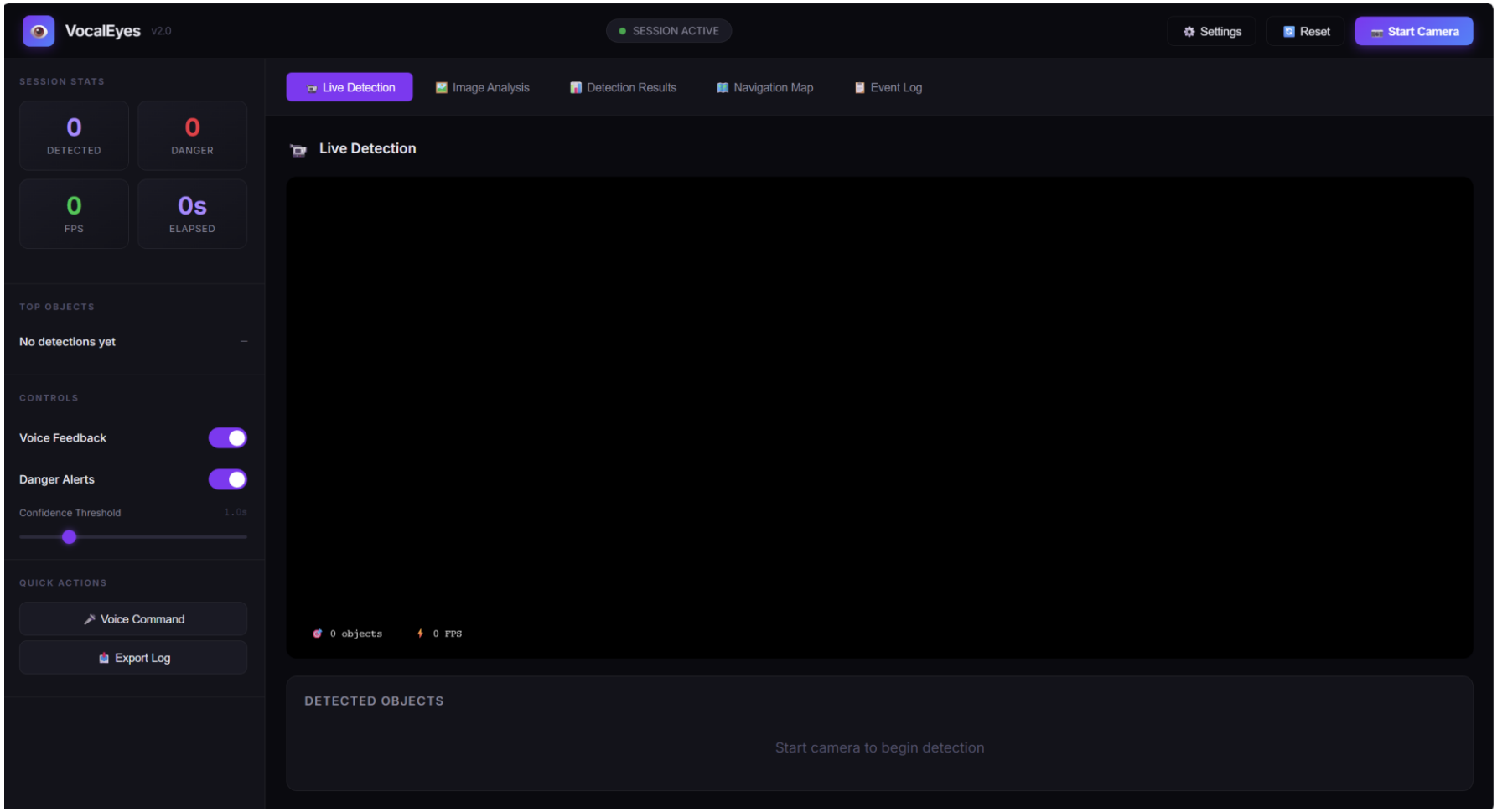


Implementation

- **Camera Input & Object Detection**
- Live video is captured using OpenCV and processed with pretrained YOLOv8 to detect people, vehicles, and common objects in real time.
- **Spatial Estimation**
- Bounding box analysis is used to estimate approximate distance (near/medium/far) and direction (left/center/right) of detected objects.
- **Priority Classification**
- Detected objects are categorized into High, Medium, and Low priority based on safety relevance, ensuring hazardous objects receive immediate attention.
- **Time-Based Intelligent Silence**
- A cooldown mechanism suppresses repeated audio alerts, reducing continuous narration and cognitive overload.

- **Conditional OCR**
- OCR is enabled only when no safety-critical objects are present, providing assistive text reading without interrupting hazard alerts.
- **Audio Feedback**
- Detected information is converted into speech using offline text-to-speech, enabling hands-free real-time guidance.
- **Event Logging System**
- Navigation Map Integration
- Settings Panel (Customization)
- **Voice Interaction**
- Supports speech commands and dynamic control of audio feedback
- **Real-Time Detection Loop**
- Continuously processes video frames at regular intervals for live monitoring

System Demonstration



- Live camera detection with bounding boxes
- Real-time audio feedback
- Dashboard showing object count and danger levels
- History tracking of detected objects
- Priority-based alerts (HIGH / MEDIUM / LOW)
- Voice command control (start/stop system)
- Interactive UI showing live detection and control options

Conclusion

- VocalEyes is a context-aware assistive vision system that integrates real-time YOLO-based object detection, spatial estimation, priority-based safety logic, time-based intelligent silence, and safety-controlled OCR to support visually impaired users.
- The system goes beyond simple object detection by combining AI, decision intelligence, user interface, and voice-based control into a unified assistive framework for real-world usability.
- Improves user safety through priority-based alerts
- Reduces cognitive overload using intelligent silence
- Enhances real-time situational awareness for better navigation

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Thank You